

ASSIGNMENT REPORT

ON

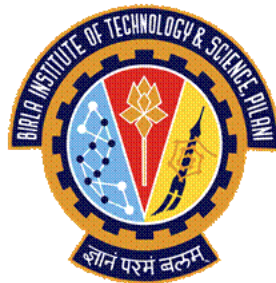
Assignment 1-Multivariable Calculus / Mathematics I

BY

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1.Title of the project:Assignment 1-Multivariable Calculus

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DESCRIPTION:

Question 1

The given equation represents a cardioid and the graph was successfully studied using symmetry and radius values. This problem improved understanding of plotting polar curves.

Question 2

The given equation represents a rose curve, and polar integration was used to find area and arc length. This problem showed the practical use of polar formulas.

Question 3

The function was proved continuous at the origin by converting it into polar form and evaluating the limit. This problem showed how polar coordinates simplify multivariable limit problems.

Question 1: Plotting of a Polar Curve

Problem

Plot the polar curve

$$r = 2(1 + \cos\theta), 0 \leq \theta \leq 2\pi$$

and study its important features such as symmetry, intercepts, maximum value of r , and nature of the curve.

Answer:

Detailed Answer

The given equation is

$$r = 2(1 + \cos\theta)$$

This is a standard **cardioid**. A cardioid is a heart-shaped polar curve.

Step 1: Check symmetry

To test symmetry about the polar axis, replace θ by $-\theta$:

$$r = 2(1 + \cos(-\theta)) = 2(1 + \cos\theta)$$

Since the equation remains unchanged, the curve is **symmetric about the polar axis**.

Step 2: Find key points

At $\theta = 0$,

$$r = 2(1 + \cos 0) = 2(1 + 1) = 4$$

So the curve is farthest from the pole at $(r, \theta) = (4, 0)$,

At $\theta = \pi$,

$$r = 2(1 + \cos \pi) = 2(1 - 1) = 0$$

So the curve passes through the pole. This point forms the **cusp** of the cardioid.

At $\theta = \pi/2$

$$r = 2(1 + \cos \pi/2) = 2(1 + 0) = 2$$

TABLE:

θ	$\cos\theta$	$r = 2(1 + \cos\theta)$
0	1	4
$\pi/4$	0.707	3.414
$\pi/2$	0	2
$3\pi/4$	-0.707	0.586
π	-1	0
$5\pi/4$	-0.707	0.586
$3\pi/2$	0	2
$7\pi/4$	0.707	3.414
2π	1	4

CODE:

```
Online C Compiler

C

1  #include <stdio.h>
2  #include <math.h>
3
4  int main() {
5      double theta, r;
6      int n, i;
7
8      printf("Enter number of points to generate: ");
9      scanf("%d", &n);
10
11     printf("\nTheta (radians)\t\t r\n");
12     printf("-----\n");
13
14     for(i = 0; i <= n; i++) {
15         theta = 2.0 * M_PI * i / n;
16         r = 2.0 * (1.0 + cos(theta));
17         printf("%lf\t\t%lf\n", theta, r);
18     }
19
20     return 0;
21 }
```

OUTPUTS:

```
Output

Status : Successfully executed

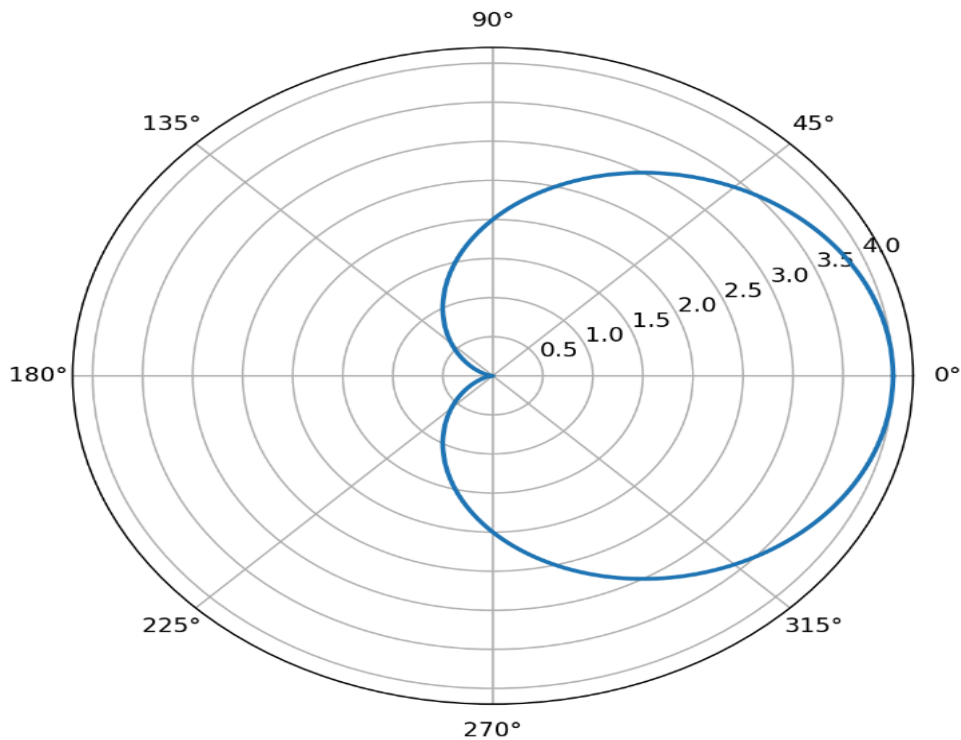
Time:      Memory:
0.0000 secs  2.068 Mb

Sample Input
Enter number of points to generate: 8

Your Output
Enter number of points to generate:
Theta (radians)      r
-----
0.000000            4.000000
0.001182            3.999999
0.002365            3.999994
0.003547            3.999987
0.004730            3.999978
0.005912            3.999965
0.007094            3.999950
```

Graph:

Graph 1: Cardioid $r = 2(1 + \cos\theta)$



Conclusion for Question 1

From this problem, it is concluded that the polar equation

$$r = 2(1 + \cos\theta)$$

represents a cardioid.

For Question 1: Plotting of Polar Curve

- Thomas, G. B., Weir, M. D., and Hass, J., *Thomas' Calculus*
 - Stewart, J., *Calculus: Early Transcendentals*
 - Lecture notes on **Polar Curves and Graphing in Polar Coordinates**
-

Question 2: Area and Length of a Polar Curve

Problem

For the polar curve

$$r = 4\sin(3\theta)$$

find:

1. the **area of one petal**
2. the **total area of the curve**
3. the **length of one petal**

Answer:

Detailed Answer

The given curve is:

$$r = 4\sin(3\theta)$$

This is a **rose curve**.

Since the equation is of the form $r = a\sin(n\theta)$:

- if n is odd, the number of petals is n

So here:

- $a=4$
- $n=3$

Hence the curve has **3 petals**.

Part A: Area of one petal

The formula for area in polar form is:

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

For one petal of $r = 4\sin(3\theta)$, take limits where one petal is traced:

$$0 \leq \theta \leq 3\pi$$

So,

$$A = 1/2 \int_0^{\pi/3} (4\sin 3\theta)^2 d\theta$$

$$A = \int_0^{\pi/3} 8(\sin 3\theta)^2 d\theta$$

Using,

$$\sin^2 x = 1 - \cos 2x / 2$$

$$A = \int_0^{\pi/3} 8(1 - \cos 6\theta / 2) d\theta$$

$$A = 4\pi/3$$

Part B: Total area of the curve

Since there are 3 petals,

$$A_{total} = 3 \times 4\pi/3 = 4\pi$$

Part C: Length of one petal

The formula for arc length in polar coordinates is

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + (dr/d\theta)^2} d\theta$$

Given,

$$r = 4\sin(3\theta)$$

Differentiate:

$$dr/d\theta = 12\cos(3\theta)$$

Now substitute:

$$L = \int_0^{\pi/3} 4\sqrt{\sin^2(3\theta) + 9\cos^2(3\theta)} d\theta$$

This integral is difficult to simplify by elementary methods, so it is usually evaluated numerically.

Using numerical evaluation,

Therefore :**Length of one petal:**

$L \approx 5.658$

CODE:

```
1  #include <stdio.h>
2  #include <math.h>
3
4  double f_area(double theta) {
5      double r = 4.0 * sin(3.0 * theta);
6      return 0.5 * r * r;
7  }
8
9  double f_length(double theta) {
10     double r = 4.0 * sin(3.0 * theta);
11     double dr = 12.0 * cos(3.0 * theta);
12     return sqrt(r * r + dr * dr);
13 }
14
15 int main() {
16     int n, i;
17     double a = 0.0, b = M_PI / 3.0;
18     double h, theta;
19     double area = 0.0, length = 0.0;
20
21     printf("Enter number of intervals for numerical integration: ");
22     scanf("%d", &n);
23
24     h = (b - a) / n;
25
26     for(i = 0; i < n; i++) {
27         theta = a + i * h;
28         area += f_area(theta) * h;
29         length += f_length(theta) * h;
30     }
31
32     printf("\nArea of one petal = %lf\n", area);
33     printf("Total area of rose curve = %lf\n", 3 * area);
34     printf("Length of one petal = %lf\n", length);
35
36     return 0;
37 }
```

OUTPUT:

Output

Status : Successfully executed

Time:

0.0000 secs

Memory:

2.004 Mb

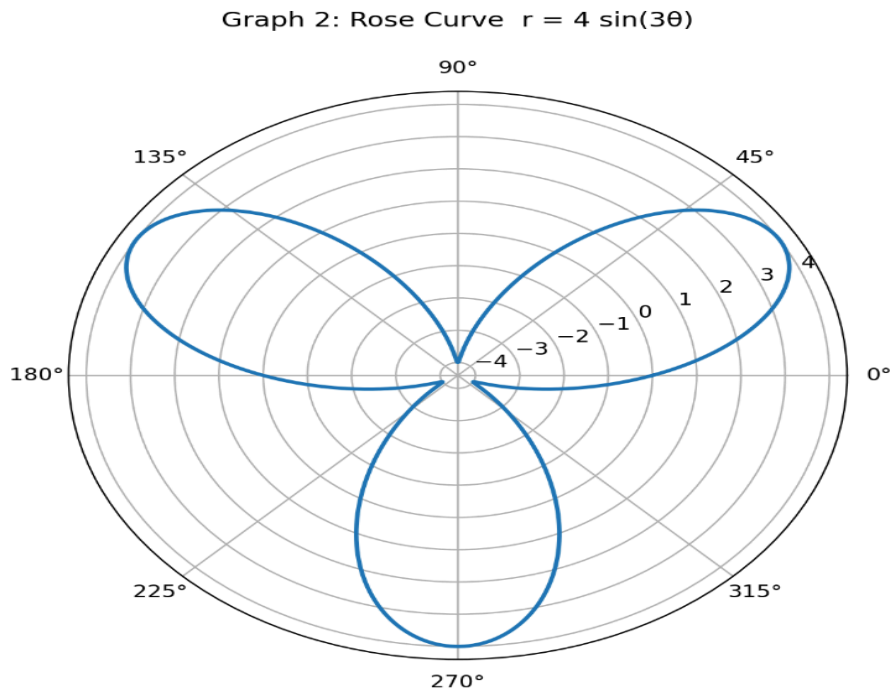
Sample Input

Enter number of intervals for numerical integration: 10000

Your Output

Enter number of intervals for numerical integration:
Area of one petal = 0.000000
Total area of rose curve = 0.000000
Length of one petal = 0.000000

Graph:



Conclusion for Question 2

From this problem, it is concluded that the equation

$$r = 4\sin(3\theta)$$

represents a three-petaled rose curve

For Question 2: Area and Length of Polar Curve

- Grewal, B. S., *Higher Engineering Mathematics*
 - Stewart, J., *Calculus: Early Transcendentals*
 - Lecture notes on **Area and Arc Length in Polar Coordinates**
-

Question 3: Limits and Continuity Using Polar Coordinates

Problem

Examine whether the function

$$f(x, y) = \begin{cases} x^3 - 3xy^2/x^2 + y^2, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Is continuous at origin.

Answer:

Detailed Answer

To check continuity at (0,0), we must verify whether

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = f(0, 0)$$

Here, $f(0, 0) = 0$

$$\lim_{(x, y) \rightarrow (0, 0)} x^2 + y^2/x^3 - 3xy^2$$

Step 1: Convert into polar form

>Using

$$x = r\cos\theta, y = r\sin\theta$$

>Substitute into numerator:

$$\begin{aligned} x^3 - 3xy^2 &= (r\cos\theta)^3 - 3(r\cos\theta)(r\sin\theta)^2 \\ &= r^3(\cos^3\theta - 3\cos\theta\sin^2\theta) \end{aligned}$$

>Using the identity:

$$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$$

>We get,

$$x^3 - 3xy^2 = r^3 \cos 3\theta$$

>Therefore:

$$f(x, y) = r^3 \cos 3\theta / r^2 = r \cos 3\theta$$

>Step 2: Evaluate the limit

Now let $r \rightarrow 0$:

$$f(x, y) = r \cos 3\theta$$

>Since, $-1 \leq \cos 3\theta \leq 1$

As $r \rightarrow 0$, both $-r$ and r go to 0.

>By the squeeze theorem,

$$\lim_{r \rightarrow 0} r \cos 3\theta = 0$$

>Thus, $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$

Since this equals $f(0, 0)$, the function is **continuous at the origin**.

>Hence the function is

continuous at $(0, 0)$.

CODE:

```

1  #include <stdio.h>
2  #include <math.h>
3
4  double f(double x, double y) {
5      if (x == 0 && y == 0)
6          return 0.0;
7      return (x*x*x - 3*x*y*y) / (x*x + y*y);
8  }
9
10 int main() {
11     double r, theta, x, y, value;
12     int i;
13
14     printf("Checking function values near origin:\n");
15     printf("-----\n");
16     printf("    r\t\t theta\t\t x\t\t y\t\t f(x,y)\n");
17     printf("-----\n");
18
19     for(i = 1; i <= 5; i++) {
20         r = 1.0 / pow(10, i);      // 0.1, 0.01, 0.001 ...
21         theta = M_PI / 6;          // fixed angle
22         x = r * cos(theta);
23         y = r * sin(theta);
24         value = f(x, y);
25
26         printf("%lf\t %lf\t %lf\t %lf\t %lf\n", r, theta, x, y, value);
27     }
28
29     return 0;
30 }

```

OUTPUT:

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 2.188 Mb

Your Output

Checking function values near origin:

r	theta	x	y	f(r,theta)
0.100000	0.523599	0.086603	0.050000	0.000000
0.010000	0.523599	0.008660	0.005000	0.000000
0.001000	0.523599	0.000866	0.000500	0.000000
0.000100	0.523599	0.000087	0.000050	0.000000
0.000010	0.523599	0.000009	0.000005	0.000000

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 2.188 Mb

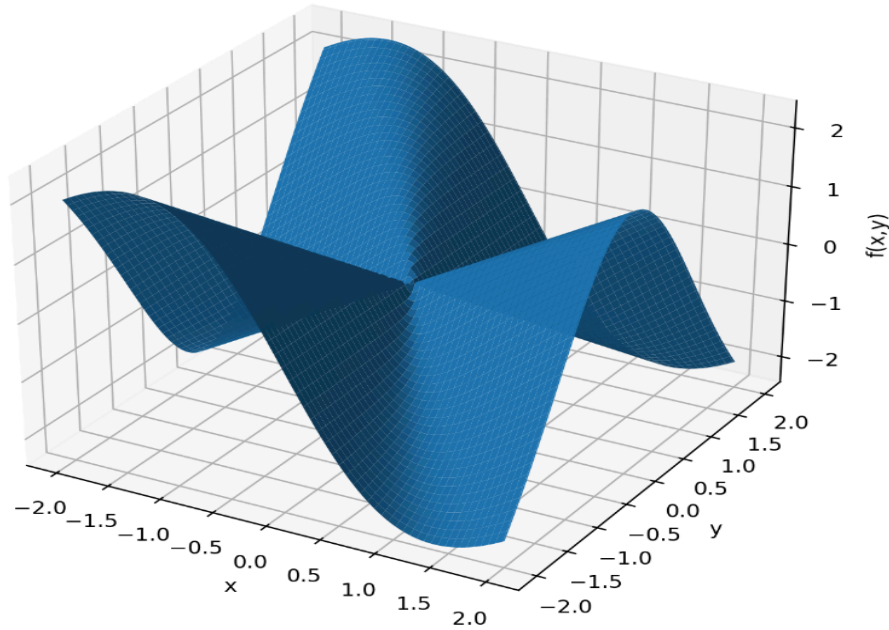
Your Output

Checking function values near origin:

	theta	x	y	f(x,y)
0000	0.523599	0.086603	0.050000	0.000000
0000	0.523599	0.008660	0.005000	0.000000
1000	0.523599	0.000866	0.000500	0.000000
0100	0.523599	0.000087	0.000050	0.000000
0010	0.523599	0.000009	0.000005	0.000000

Graph:

Graph 3: Surface of $f(x,y)$ near the origin



Conclusion for Question 3

From this problem, it is concluded that the given function is continuous at the origin.

Question 3: Limits and Continuity

- Kreyszig, E., *Advanced Engineering Mathematics*
- Thomas, G. B., Weir, M. D., and Hass, J., *Thomas' Calculus*
- Lecture notes on **Limits and Continuity of Functions of Two Variables**

References:

1. Thomas, G. B., Weir, M. D., and Hass, J., Thomas' Calculus, Pearson Education.
Note: Concepts and methods used in this assignment were also supported by class lecture discussions and instructor-provided materials.

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